Problem Set
Application of Support Vector Machines

October 26, 2025

1. Load the breast cancer dataset from `sklearn` library. Identify the response (target classes, malignant or benign) and the predictors.
2. Standardize the predictors, if needed.
3. Split the dataset into training and test parts in $75\% - 25\%$ ratio.
4. Fit a support vector classifier to the data with the linear kernel and the default choice of the regularization parameter C, and print the Training and test accuracy.
5. Try the same with different other values of the regularization parameter.
6. Choose an appropriate value of C with a 5-fold CV on the training data.
7. Fit a support vector classifier to the data with the radial basis function kernel. From a set of different values of regularization parameter C and kernel coefficient γ, choose the best combination using a 5-fold CV.
8. Try step 7 with a polynomial kernel.
9. Fit a logistic regression on the training data. Compute the training and test error rates.
10. Based on training and test error, which classifier do you prescribe among those used in this problem set?

Optional Task

- Apply Recursive Feature Elimination (RFE) using SVM to identify the most important subset of features.
- Analyze how reducing the number of features affects model performance.